

Shah zaman (097)

Question 1

Assign 0 to x if $y = 10$, otherwise 1.

```
#include <iostream>
using namespace std;
```

```
int main() {
    int x, y;
    cin >> y;

    if (y == 10)
        x = 0;
    else
        x = 1;

    cout << "x = " << x;

    return 0;
```

```
}
```

Question 2

Print message if temperature is between -50 and 150.

```
#include <iostream>
using namespace std;
```

```
int main() {
    int temperature;

    cin >> temperature;

    if (temperature >= -50 && temperature <=
150) {
        cout << "The number is valid";
    }

    return 0;
}
```

Question 3

Program to find smaller and larger number.

```
#include <iostream>
using namespace std;
```

```
int main() {
    int num1, num2;

    cout << "Enter two numbers: ";
    cin >> num1 >> num2;

    if (num1 > num2) {
        cout << "Larger number: " << num1 <<
endl;
        cout << "Smaller number: " << num2;
    } else {
        cout << "Larger number: " << num2 <<
endl;
        cout << "Smaller number: " << num1;
    }
}
```

```
    return 0;
}
```

Question 4

Add 0.2 to a and compare.

```
#include <iostream>
using namespace std;
```

```
int main() {
    double a = 10;
    double b = 10;

    a = a + 0.2;

    if (a > b)
        cout << "a is larger\nb is smaller";
    else
        cout << "b is larger\na is smaller";

    return 0;
}
```

Question 5

Division only if denominator $\neq 0$.

```
#include <iostream>
using namespace std;

int main() {
    double num1, num2;

    cout << "Enter two numbers: ";
    cin >> num1 >> num2;

    if (num2 != 0)
        cout << "Division = " << num1 / num2;
    else
        cout << "Division by zero is not possible.";

    return 0;
}
```

Question 6

Given $a = 2$, $b = 4$, $c = 6$

Condition Result

A) $a == 4 \parallel b > 2 \rightarrow \text{True}$

B) $6 \leq c \ \&\& \ a < 3 \rightarrow \text{True}$

C) $1 \neq b \ \&\& \ c \neq 3 \rightarrow \text{True}$

D) $a \geq -1 \parallel a \leq b \rightarrow \text{True}$

E) $!(a > 2) \rightarrow \text{True}$

Question 7

Race result for 3 runners.

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int r1, r2, r3;
```

```
    cout << "Enter time of runner 1: ";
```

```
    cin >> r1;
```

```
    cout << "Enter time of runner 2: ";
```

```
    cin >> r2;
```

```
cout << "Enter time of runner 3: ";  
cin >> r3;
```

```
if (r1 < r2 && r1 < r3)  
    cout << "Runner 1 came first";  
else if (r2 < r1 && r2 < r3)  
    cout << "Runner 2 came first";  
else  
    cout << "Runner 3 came first";
```

```
return 0;
```

```
}
```

Question 8

Geometry Calculator

```
#include <iostream>  
using namespace std;
```

```
int main() {  
    int choice;  
    double radius, length, width, base, height, area;
```

```
cout << "Geometry Calculator\n";  
cout << "1. Area of Circle\n";  
cout << "2. Area of Rectangle\n";  
cout << "3. Area of Triangle\n";  
cout << "Enter your choice: ";  
cin >> choice;
```

```
if (choice == 1) {  
    cout << "Enter radius: ";  
    cin >> radius;  
    area = 3.14159 * radius * radius;  
    cout << "Area = " << area;  
}  
else if (choice == 2) {  
    cout << "Enter length and width: ";  
    cin >> length >> width;  
    area = length * width;  
    cout << "Area = " << area;  
}  
else if (choice == 3) {  
    cout << "Enter base and height: ";
```



```
    cin >> base >> height;  
    area = 0.5 * base * height;  
    cout << "Area = " << area;  
}
```

```
    return 0;  
}
```

Question 9

Display ASCII values of alphabets.

```
#include <iostream>  
using namespace std;
```

```
int main() {  
  
    for(char ch='A'; ch<='Z'; ch++) {  
        cout << ch << " = " << (int)ch << endl;  
    }  
  
    return 0;  
}
```

Question 10

Calculate $\text{Weight} = \text{mass} \times 9.8$

```
#include <iostream>
using namespace std;
```

```
int main() {
    double mass, weight;

    cout << "Enter mass: ";
    cin >> mass;

    weight = mass * 9.8;

    cout << "Weight = " << weight << "
    Newtons\n";

    if (weight > 1000)
        cout << "Object is too heavy";
    else if (weight < 10)
        cout << "Object is too light";
```

```
    return 0;  
}
```

Question 11

Sales bonus condition.

```
#include <iostream>  
using namespace std;  
  
int main() {  
    double sales;  
  
    cout << "Enter sales: ";  
    cin >> sales;  
  
    if (sales > 50000) {  
        sales = sales + (sales * 0.25) + 250;  
    }  
  
    cout << "Total sales = " << sales;
```

```
    return 0;
}
```

Question 12

Loan qualification program.

```
#include <iostream>
using namespace std;
```

```
int main() {
    double income;
    int years;
```

```
    cout << "Enter annual income: ";
    cin >> income;
```

```
    cout << "Enter years of employment: ";
    cin >> years;
```

```
    if (income >= 35000 && years > 5)
        cout << "You qualify for the loan";
    else
```

```
cout << "You do not qualify";
```

```
return 0;
```

```
}
```

Question 13

TV model features.

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {  
    int model;
```

```
    cout << "Enter TV Model (100,200,300): ";  
    cin >> model;
```

```
    if (model == 300)  
        cout << "Features: Picture-in-picture, Stereo  
sound, Remote control";  
    else if (model == 200)
```

```
        cout << "Features: Stereo sound, Remote  
control";  
    else if (model == 100)  
        cout << "Features: Remote control only";  
    else  
        cout << "Invalid Model";  
  
    return 0;  
}
```

Question 14

Magic Date Program.

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int month, day, year;  
  
    cout << "Enter month: ";  
    cin >> month;
```

```
cout << "Enter day: ";  
cin >> day;
```

```
cout << "Enter year: ";  
cin >> year;
```

```
if (month * day == year)  
    cout << "Date is MAGIC";  
else  
    cout << "Date is NOT MAGIC";
```

```
return 0;
```

```
}
```

Question 15

Commission calculation.

```
#include <iostream>  
using namespace std;
```

```
int main() {  
    double sales, commission;
```

```
cout << "Enter sales amount: ";  
cin >> sales;
```

```
if (sales <= 10000)  
    commission = sales * 0.10;  
else if (sales <= 15000)  
    commission = sales * 0.15;  
else  
    commission = sales * 0.20;
```

```
cout << "Commission = " << commission;
```

```
return 0;
```

```
}
```


Question no:- 16.

```
#include <iostream>
using namespace std;
int main() { double balance, fee = 10;
    int checks;
    cout << "Enter beginning balance:";
    cin >> balance;
    cout << "Enter number of
        checks written: ";
    cin >> checks;
    cout << "Enter
        if (balance < 400)
            fee += 15;
        else if (checks <= 39)
            fee += checks * 0.10;
        else if (checks < 20)
            fee += checks * 0.08;
        else if (checks <= 59)
            fee += checks * 0.06;
        else
            fee += checks * 0.04;
    cout << "Bank Service Fee for the month"
```



```
<< Fee ;
```

```
return 0 ;
```

```
}
```

Question no: 17

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int Choice ;
```

```
    cout << "Select Headphone Type \n" ;
```

```
    cout << "1. Noise Chanceling \n" ;
```

```
    cout << "2. wireless \n" ;
```

```
    cout << "3. wired budget \n" ;
```

```
    cout << "Enter your Choice : " ;
```

```
    cin >> Choice ;
```

```
    if (Choice == 1)
```

```
        cout << "Price = $ 249.00" ;
```

```
    else if (Choice == 2)
```

```
        cout << "Price = $ 199.00" ;
```

```
    else if (Choice == 3)
```

```
        cout << "Price = $ 49.00" ;
```

```
    else
```

```
        cout << "invalid Choice" ;
```

```
    return 0 ;
```

```
}
```


Question no: 18.

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {  
    int choice, months;
```

```
    double rate, total;
```

```
    cout << "1. Standard Adult Membership\n";
```

```
    cout << "2. Child Membership\n";
```

```
    cout << "3. Senior citizen Membership\n";
```

```
    cout << "4. Quit\n";
```

```
    cout << "Enter your choice:";
```

```
    cin >> choice;
```

```
    if (choice == 4)
```

```
        return 0;
```

```
    cout << "Enter number of months:";
```

```
    cin >> months;
```

```
    if (choice == 1)
```

```
        rate = 40;
```

```
    else if (choice == 2)
```

```
        rate = 20;
```

```
    else if (choice == 3)
```

```
        rate = 30;
```

```
    else {
```

```
        cout << "Invalid choice";
```

```

        return 0;
    }
    total = rate * months;
    cout << "total amount to pay: Rs " << total;
    return 0;
}

```

Question No 19

```

#include <iostream>
using namespace std;
int main() {
    int package;
    double gb, bill;
    cout << "Enter package (1, 2, 3): ";
    cin >> package;
    cout << "Enter gigabytes used: ";
    cin >> gb;
    if (package == 1) {
        bill = 39.99;
        if (gb > 4)
            bill += (gb - 4) * 10;
    }
    else if (package == 2) {
        bill = 59.99;
        if (gb > 4)

```



```

    bill += (gb - 8) * 10 * 5;
}
else if (package == 3) {
    bill = 69.99;
}
else {
    cout << "Invalid Package";
    return 0;
}
cout << "Total amount due $" << bill;
return 0;
}

```

Question No 20

```

#include <iostream>
using namespace std;
int main () {
    int Seconds;
    cout << "Enter number of Seconds: ";
    cin >> Seconds;
    if (Seconds >= 86400)
        cout << "Day: " << Seconds / 86400 << endl;
    if (Seconds <= 3600)
        cout >> "Hours: " << Seconds / 3600 << endl;
}

```

if (Seconds $\geq$ 60)

cout << "Minutes: " << Seconds / 60 << endl;

return 0;

}